State inputs to the ECEC simulator

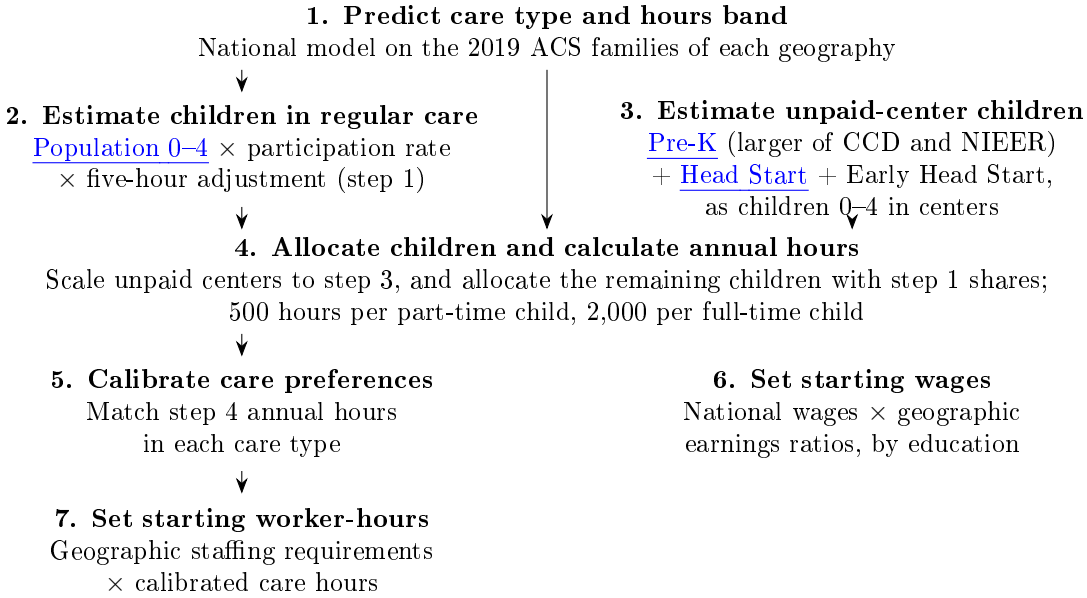
Anna Rachwalski

8 October 2026

This memo updates the 30 September memo on the state inputs to the ECEC simulator. These are the changes since then.

- The care mix of each geography now comes from predictions for its ACS families.
- Participation pools the 2016–19 NSCH rates, and the five-hour adjustment is specific to each geography.
- I revised the public-program counts and use one Head Start overlap share, 27.5%, everywhere.
- Wages, staffing, and non-labor costs vary by geography, and NYC has its own run.
- Family prices use income ranks within the state. Caregiver employment follows ACS trends by sex and age and national projections, in place of local unemployment trends.

The base year is 2019, and demand covers children ages 0–4. NYS includes NYC.



1. Demand (steps 1–5)

1.1 Predicted care (step 1)

I apply the national model before geographic calibration to the 2019 ACS families of each geography. It predicts children by care type and hours band. Part-time is under 20 hours a week in the main arrangement, and the model counts each part-time child at 10 hours a week.

Before prediction, I adjust child weights for national Census–ACS differences in single-year age shares.

$$\text{adjusted child weight} = \text{ACS weight} \times \frac{\text{national Census share of children 0–4 at the child's age}}{\text{national ACS weighted share of children 0–4 at the child's age}},$$

$$\text{predicted part-time share} = \frac{\text{predicted part-time children}}{\text{predicted children in care}}.$$

1.2 Children in care (step 2)

The NSCH participation rate is the share of children 0–4 in non-parental care for at least 10 hours a week, from the National Survey of Children's Health (NSCH, item K6Q20). NYC takes the NYS rate.

I pool the 2016–19 NSCH records with child weights and keep each wave's survey design. Against 2019 alone, the median state SE falls from 5.8 to 2.8 percentage points.

The national five-hour adjustment is 1.098: children 0–4 in regular care for at least five hours a week divided by children in care for at least ten, in the National Survey of Early Care and Education (NSECE). I adjust it for each geography with the predicted part-time share: more part-time children means more children in the 5–10 hour range the NSCH rate misses.

Among NSECE children whose main arrangement is part-time, 34.0% get at least five but fewer than ten hours a week of care in total, over all arrangements. I assume this applies in every geography.

$$\text{relative factor} = \frac{0.66 \times \text{national predicted part-time share}}{0.66 \times \text{predicted part-time share of the geography}},$$

$$\text{children in care} = \text{population 0–4} \times \text{NSCH participation rate} \times 1.098 \times \text{relative factor}.$$

1.3 Children in public programs (step 3)

Unpaid center is center care the family doesn't pay for: public pre-K, Head Start, and Early Head Start.

For pre-K, I use the larger of public-school enrollment from the Common Core of Data (CCD) and state-funded enrollment for ages 3 and 4 in all settings from the National Institute for Early Education Research (NIEER). I exclude the special-education count NIEER adds.

NYC uses the city enrollment in pre-K and 3-K, ages 3 and 4, in all settings, October 2018.

Head Start and Early Head Start counts are funded slots, from NIEER. NYC takes the NYS counts times the city's share of funded slots in the Head Start service-location file.

The coefficients in the equation below adjust program counts for enrollment, center setting, and age (0–4 on 31 December), with the 2018–19 Head Start Program Information Report and age shares on 31 December 2018 from the National Household Education Surveys (NHES). NYC enrolls pre-K by calendar year, and NYC Head Start follows the same rule.

The Head Start overlap share is the share of Head Start slots the pre-K count already includes. I use one share in every geography, from the geographies where I use the NIEER pre-K count and NIEER reports Head Start without duplication. I assume their overlap is representative.

$$\text{NIEER overlap share} = 1 - \frac{\text{Head Start children NIEER adds to its pre-K and special-education total}}{\text{Head Start slots}},$$

$$\text{Head Start overlap share} = \frac{\text{total over the ten geographies of Head Start slots} \times \text{NIEER overlap share}}{\text{total over the ten geographies of Head Start slots}} = 0.275.$$

$$\begin{aligned}\text{Unpaid center children} = {} & 0.756 \times \text{pre-K enrollment} \\ & + 0.838 \times \text{Head Start slots} \times (1 - \text{Head Start overlap share}) \\ & + 0.551 \times \text{Early Head Start slots}.\end{aligned}$$

This equation applies outside NYC. NYS is the sum of the NYC result and the rest of the state with the national coefficients. I use these public-program counts as the unpaid-center target.

1.4 Remaining children and hours (steps 4–5)

I scale the step 1 predictions in two groups: Unpaid center, and the other five types.

$$\text{Unpaid center factor} = \frac{\text{Unpaid center children (step 3)}}{\text{predicted Unpaid center children}},$$

$$\text{other-care factor} = \frac{\text{children in care (step 2)} - \text{Unpaid center children (step 3)}}{\text{predicted children in the other five types}},$$

$$\text{annual hours} = (10 \times \text{part-time children} + 40 \times \text{full-time children}) \times 50 \text{ weeks}.$$

I apply each group's factor to both hours bands to keep each type's part-time share and the relative shares of the other five types.

I calibrate preferences to total annual hours in each of the six care types. I don't separately target part-time shares. The split between low-price and high-price centers stays as the model predicts.

Annual demand targets, 2019 (millions of child-hours)					
Care type	KY	NJ	NYS	NYC	VA
Paid center	68.4	160.1	315.5	136.0	166.5
Unpaid center	48.9	68.4	213.7	130.9	54.0
Paid home	18.3	56.4	119.3	50.3	46.1
Other paid	53.9	90.0	182.0	77.9	122.1
Unpaid home	5.0	8.8	16.1	7.1	6.0
Other unpaid	71.9	161.1	346.1	154.0	122.1

2. Supply (steps 6–7)

2.1 Wages

I set wages for two education groups: below a bachelor's degree (BA), and BA or higher.

Childcare workers are the workers in the childcare occupation outside private households, plus preschool teachers, special education teachers, and administrators in child day care services.

The sample is the 2018 and 2019 records of the 2015–19 ACS file. Hourly earnings are wage and self-employment earnings divided by weeks worked times usual weekly hours.

$$\text{geographic earnings ratio} = \frac{\text{geographic weighted median hourly earnings, childcare workers}}{\text{national weighted median hourly earnings, childcare workers}},$$

$$\text{starting wage} = \text{national NSECE wage} \times \text{geographic earnings ratio}.$$

The national NSECE wage is the simulator's 2019 wage for each education group, from the NSECE workforce data. A group with fewer than 30 unweighted ACS records uses the geographic-to-national earnings ratio of all workers in the same education group. This applies to AK, DC, DE, ID, MT, ND, NM, WV, and WY, all in the BA-or-higher group (9 of 104 geography-by-education cells).

Starting wage (2019 $/hour)		
	Below BA	BA or higher
KY	10.05	15.85
NJ	14.06	22.38
NYS	13.80	22.17
NYC	14.38	27.95
VA	11.72	21.74
United States (national NSECE wage)	12.06	19.70

2.2 Staffing and non-labor costs

For centers, required staff per child is the licensed staff-to-child ratio from the 2014 federal licensing study, averaged over ages 0–4 with NSECE center enrollment weights. NYC centers use the city Health Code, and NYC homes use the state rule.

For homes, required staff per child is 1 divided by the maximum licensed capacity of a family child care home, from the Prenatal-to-3 Policy Impact Center, Table 6. The national home value includes

only states with licensing values. States that don't license homes (AZ, ID, IN, LA, NJ, OH, and SD) keep the national home worker-hours per child-hour of the simulator.

National required staff per child is the mean of state values, weighted by population ages 0–4.

Regional Price Parity is the 2019 all-items price level of the Bureau of Economic Analysis (national = 100). NYC uses its metro area.

$$\text{worker-hours per child-hour} = \text{national worker-hours per child-hour} \times \frac{\text{state required staff per child}}{\text{national required staff per child}},$$

$$\text{non-labor costs and margin} = \text{national non-labor costs and margin} \times \frac{\text{Regional Price Parity}}{100}.$$

Licensing requirements give an approximate staffing need.

	Children per staff member (equivalent)		Non-labor costs and margin (2019 dollars per child-hour)		
	Center	Home	Low-price center	High-price center	Paid home
KY	10.5	6	0.43	4.23	2.53
NJ	10.2	–	0.53	5.25	3.14
NYS	6.1	6	0.52	5.17	3.09
NYC	7.9	6	0.55	5.43	3.25
VA	9.1	12	0.48	4.73	2.83
United States	9.1	6.8	0.48	4.72	2.83

Equivalent children per staff member is 1 divided by required staff per child. Low-price and high-price centers are the model's two center price tiers. NJ doesn't license homes. NYS uses state rules and prices everywhere in the state. NYC uses city center rules and metro prices.

2.3 Family prices

I retrain the simulator's price forest without region, with income rank, number of parents, and child age. It trains on the net hourly payment of each NSECE 2019 family divided by the national 2019 supply price of its care type.

$$\text{family price} = \text{state market price} \times \text{price multiplier}.$$

As NSECE gives no state, I rank NSECE incomes within region. I rank ACS incomes within state, and NYC within NYS.

2.4 Labor

For each education group, I sum labor over unpaid centers, low-price and high-price paid centers, and paid homes. I keep the national wage-response rule and supply elasticities.

$$\text{starting worker-hours} = \text{total over sectors of } \big(\text{calibrated child-hours} \times \text{worker-hours per child-hour}\big).$$

3. Employment (step 8)

I keep the model's 2019 caregiver employment rates and apply the employment growth of the wider population in eight sex-and-age groups: under 25, 25–34, 35–44, and 45+.

I count one primary caregiver in each family with children ages 0–4. This is the mother in two-parent families, and otherwise the first-listed parent.

The model baseline employment rate is the 2019 ACS-weighted mean predicted probability of work among the caregivers of the geography in each group.

I calculate trends in 20 narrow cells: sex by age 16–17, 18–19, 20–24, and five-year bands from 25 to 59. I combine the 20 cells into the eight groups with fixed 2019 geographic age weights, pooled across sexes. The 45+ trends use ages 45–59.

ACS employment rates cover all persons in each cell, with ACS person weights, for 2007 to 2024. Each year is pooled with the year before.

The national employment rate comes from the Current Population Survey (CPS) through 2025. From 2026, it is the labor force participation projections of the Congressional Budget Office (CBO), times the 2015–2019 mean ratio of employment rate to participation rate in each cell.

$$\begin{aligned}\text{employment trend of a cell} &= \text{national employment rate of the cell} \\ &\quad \times \frac{\text{geographic ACS employment rate of the cell}}{\text{national ACS employment rate of the cell}},\end{aligned}$$

$$\begin{aligned}\text{target caregiver employment rate} &= \text{model baseline employment rate of the group} \\ &\quad \times \frac{\text{employment trend of the group in the year}}{\text{employment trend of the group in 2019}}.\end{aligned}$$

After 2024, I freeze each cell's geographic-to-national employment ratio.

4. Next steps

I am comparing the paid-center and paid-home targets with enrollment estimates from the licensing registers. These checks will inform revisions to the targets. Overall, the targets have 15.5% fewer children in licensed centers than the registers, and 27.9% fewer in licensed homes. If I set paid centers and paid homes from the registers, the targets change as follows.

- Paid center children increase by 23.9% overall.
- Paid home children increase by 36.2% overall.
- Other paid, other unpaid, and unpaid home children decrease by 18.4% overall.